Predicting GDP per Capita Across Income Groups

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Abstract

This paper investigates the extent to which cross-country differences in GDP per capita can be explained and predicted by variables suggested by the augmented Solow growth model. I construct a panel dataset for 217 economies over 1990–2023 by merging World Bank World Development Indicators with the Penn World Table, classify countries into low-, middle- and high-income groups, and estimate a hierachy of models of increasing flexibility. Starting from a benchmark Solow regression in logs, I introduce regularisation (LASSO and ridge regression), extensive feature engineering (lags, interactions and non-linear terms) and non-linear prediction methods (Random Forests and k-Nearest Neighbours). Out-of-sample performance is evaluated using a common train–test split. The results show that Solow-type variables are highly informative about the level of income: regularised and non-linear models substantially improve prediction relative to the baseline OLS specification, and across all approaches human capital and total factor productivity emerge as the dominant predictors of GDP per capita. Models with extensive feature engineering - especially lag structures - approach near-perfect predictive performance, reflecting the strong temporal persistence of GDP per capita.

1 Introduction

Why are some countries rich while others remain poor? Since Solow (1956), a large literature has stressed the role of capital accumulation, population growth and technology in shaping long-run income differences. Empirical work such as Mankiw et al. (1992) has shown that an augmented Solow model, in which human capital complements physical capital, can account for a sizable fraction of cross-country variation in output per worker.

This paper revisits these themes using a modern data-science toolkit. Concretely, I address the following question:

How accurately can we predict GDP per capita using variables motivated by the augmented Solow model, and how does predictive performance vary across low-, middle- and high-income economies?

To answer this, I proceed in four steps. First, I construct a country-year panel by merging macroeconomic series from the World Bank with human capital and productivity indices from the Penn World Table. Second, I classify countries into income groups using a GNI-per-capita rule inspired by the World Bank Atlas method. Third, I estimate a baseline Solow regression for log GDP per capita and then extend it with feature engineering (lags, interactions and non-linear transformations). Finally, I compare a range of predictive models - OLS, penalised regressions, Random Forests and k-Nearest Neighbours - on a common train-test split.

The contribution of the paper is deliberately modest: the objective is not to provide structural causal estimates, but to assess how far Solow-type variables can go in predicting the level of income when combined with modern regularisation and machine-learning techniques. The analysis confirms the central role of human capital and total factor productivity, while highlighting the importance of non-linearities and heterogeneity across income groups.

This project also allowed me to connect the theoretical Solow framework studied in *Quantitative and Statistical Methods I* and *Economic Growth and Development* to real-world data through a fully empirical implementation.

The remainder of the paper is organised as follows. Section 2 describes the construction of the dataset and the main variables. Section 3 discusses data cleaning and the income-group classification. Section 4 outlines the feature engineering strategy. Section 5 presents the empirical methodology and results. Section 6 concludes.

2 Data

2.1 Sources and coverage

The empirical analysis combines three main sources of information:

(i) **World Bank World Development Indicators (WDI).** I use the following series:

- GDP, constant 2015 US$ (code `NY.GDP.MKTP.KD`);
- GNI, Atlas method, current US$ (`NY.GNP.ATLS.CD`);
- population, total (`SP.POP.TOTL`);
- labour force, total (`SL.TLF.TOTL.IN`);
- gross savings, current US$ (`NY.GNS.ICTR.CD`);

- net primary income from abroad, current US$ (`NY.GSR.NFCY.CD`).

(ii) **Penn World Table.** From the Penn World Table I extract:

- a human capital index (`hc`);
- TFP at constant national prices (`rtfpna`), interpreted as a measure of technology A_t.

(iii) **World Bank metadata.** Using the `WDI` package in `R`, I obtain a country-level region classification and a flag identifying aggregate entities (such as "Euro area" or "World"), which are later excluded from the analysis.

The WDI series are provided in wide format with one column per calendar year. In the empirical implementation, they are reshaped into long format with variables `Country.Name`, `Country.Code`, `year` and a single value column for each indicator. The Penn World Table extract undergoes a similar transformation, yielding country-year series for human capital and TFP. Merging on the ISO3 country code and calendar year produces a merged dataset that spans 1960-2024 and includes 266 entities (countries plus aggregates) prior to filtering.

2.2 Solow variables and transformations

The empirical model is grounded in an augmented Solow framework, in which the log of GDP per capita in country i at time t is related to the savings rate, population growth, human capital and TFP:

$$\log y_{it} = \beta_0 + \beta_1 s_{it} + \beta_2 n_{it} + \beta_3 \log h_{it} + \beta_4 \log A_{it} + \varepsilon_{it}. \tag{1}$$

To operationalise this, I construct the variables listed in Table 1 from the merged dataset.

Table 1: Mapping of Solow model variables to data

Symbol	Interpretation	Empirical counterpart
y_{it}	GDP per capita	$\log(\text{GDP}_{it}/\text{POP}_{it})$ (`log_gdp_pc`)
s_{it}	savings rate	$\text{GROSS_SAV}_{it}/\text{GDP}_{it}$ (`savings_rate`)
n_{it}	population growth	$(\text{POP}_{it} - \text{POP}_{i,t-1})/\text{POP}_{i,t-1}$ (`pop_growth`)
h_{it}	human capital	$\log(\text{human capital index}_{it})$ (`log_hc`)
A_{it}	TFP / technology	$\log(\text{TFP index}_{it})$ (`log_A_t`)
	labour-force participation	$\text{LABOUR_FORCE}_{it}/\text{POP}_{it}$ (`labour_participation`)
	net primary income share	$\text{NET_INC}_{it}/\text{GDP}_{it}$ (`net_income_share`)

In what follows, I treat log GDP per capita as the dependent variable and $(s_{it}, n_{it}, \log h_{it}, \log A_{it})$ as core regressors. Labour-force participation and the net primary income share are included as additional controls in some specifications. All levels are measured in constant or current US dollars as appropriate, and ratios are constructed at the country-year level.

3 Cleaning, sample selection and income-group classification

3.1 Removing aggregate entities

The merged dataset includes both sovereign countries and aggregate entities (e.g. "Arab World", "Euro area"). Including aggregates risks double-counting populations and distorting cross-country comparisons. I therefore use the World Bank metadata to flag observations with `wb_region == "Aggregates"` and remove them from the panel. This step excludes 49 aggregates and reduces the sample to 217 individual economies and roughly 14,000 country-year observations.

3.2 Missing data diagnostics and time restriction

Before estimation, it is important to understand the structure of missing values in the key variables. Figure 1 plots, for each variable, the count of missing observations in the full merged dataset.

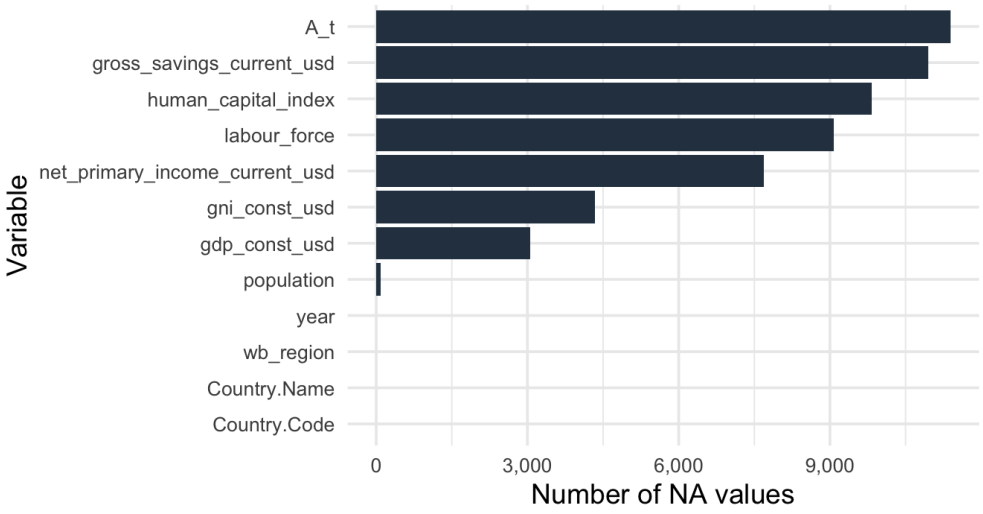


Figure 1: Number of missing values by variable

Human capital and TFP exhibit particularly severe missingness in earlier decades, while GDP and population are relatively well covered. To assess temporal coverage, I compute, for each year, the share of countries for which *all* core Solow regressors are observed. The resulting series is displayed in Figure 2.

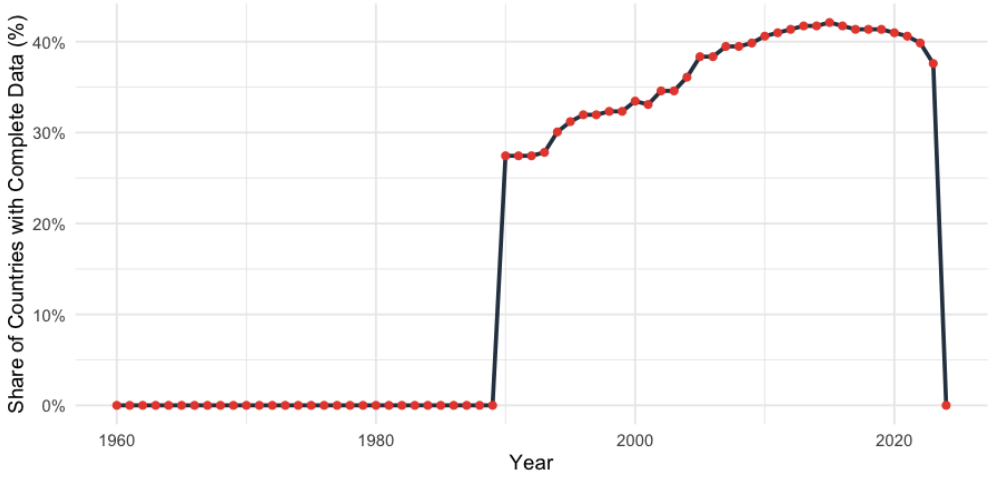


Figure 2: Share of countries with complete Solow data, by year

Completeness is very low before 1990, with fewer than 10% of countries having full information. From 1990 onwards it stabilises at a markedly higher level. In order to work with a reasonably informative panel and minimise the influence of structural breaks in data availability, I therefore restrict the analysis to the period 1990–2023. This restriction balances cross-sectional coverage with temporal completeness, ensuring that missingness does not systematically distrot the regression sample. Within this window, I drop observations with non-positive GDP or population, as they are incompatible with log transformations.

3.3 Country-level completeness filter

Even after the time restriction, some countries remain extremely sparsely observed in the Solow variables. For each country I compute a completeness measure, defined as the fraction of non-missing entries among:

$$\{\texttt{log_gdp_pc}, \texttt{savings_rate}, \texttt{pop_growth}, \texttt{log_hc}, \texttt{log_A_t}, \texttt{labour_participation}, \texttt{net_income_share}\}.$$

Countries with completeness below 25% are excluded from the modelling sample. This simple rule is designed to strike a balance between retaining as many economies as possible and avoiding results driven by extremely sparse series.

After applying all filters and dropping residual missing values in the regression variables, the main modelling dataset contains approximately 3,300 country–year observations.[1]

3.4 Construction of income groups

To examine heterogeneity across levels of development, I classify countries into three income groups using a per-capita income concept inspired by the World Bank Atlas method. Let $\tilde{y}_i$ denote the per-capita income used for classification. It is constructed according to the following priority rule:

(a) If GNI (Atlas method, current US$) and population are both observed in 2023, set $\tilde{y}_i$ equal to GNI per capita in 2023.

(b) If GNI is missing but GDP and population are available in 2023, use GDP per capita in 2023.

(c) If neither GNI nor GDP is available in 2023, use the last available pre-2023 GNI per capita.

(d) If no GNI or GDP data are ever observed, set $\tilde{y}_i = 0$ and classify the country as low income. This case applies to a small number of territories with particularly poor data.

Countries are then classified into three groups based on $\tilde{y}_i$:

- low income: $\tilde{y}_i \leq \$1{,}035$;
- middle income: $1{,}036 \leq \tilde{y}_i \leq \$12{,}615$;
- high income: $\tilde{y}_i > \$12{,}615$.

Table 2 reports the resulting distribution of countries and world population by income group in 2023.

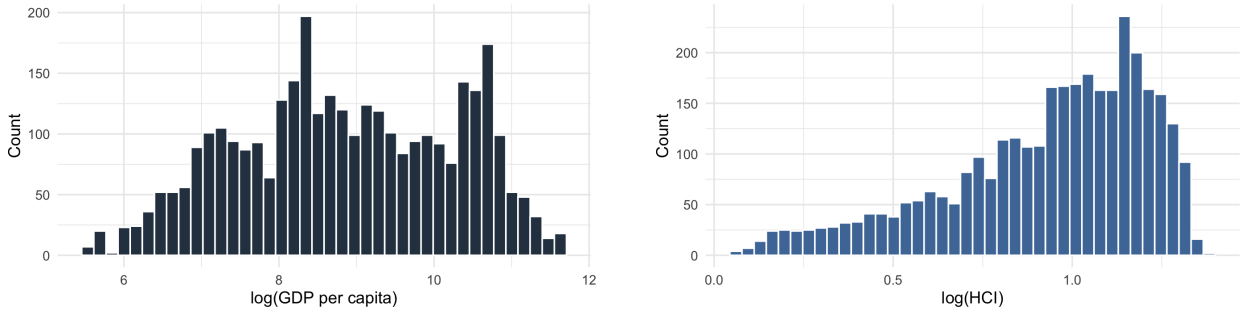
Table 2: Countries and population by income group, 2023

Income group	Number of countries	World population share	Example economies
Low income	32	725,452,079	e.g. Niger, Malawi
Middle income	103	4,446,679,741	e.g. India, Brazil
High income	82	2,869,471,498	e.g. Germany, Japan

[1]The exact number depends on the availability of PWT data for human capital and TFP.

Middle-income economies account for around 60% of world population, reflecting the demographic weight of India, China and other emerging markets.

Boxplots of log GDP per capita and the human capital index by income group, shown in Figure 3, display a pronounced monotonic gradient: high-income economies exhibit the highest levels of human capital and income, followed by middle-income countries, while low-income countries cluster at the bottom of the distributions.



(a) log GDP per capita

(b) Human capital index

Figure 3: Distributions of income and human capital by income group

These patterns motivate the interaction terms between income-group dummies and Solow variables introduced in Section 4.

4 Feature engineering

The basic specification in equation (1) is linear and static. To better capture dynamics, non-linearities and heterogeneity across income levels, I augment the feature set along three dimensions after which the engineered design matrix contains 24 predictors.

4.1 Autoregressive terms

First, I allow for persistence in output by including lagged values of log GDP per capita. In the main feature-engineered models I include up to ten lags:

$$\log y_{i,t-1}, \log y_{i,t-2}, \ldots, \log y_{i,t-10},$$

constructed within each country so that lags never cross country borders. These terms capture gradual convergence towards the steady state and other dynamic adjustment effects, but do not have structural interpretation. They are included solely to provide autoregressive persistence in the level of income.

For robustness, I also estimate an extended "AR Solow" model with twenty lags of log GDP per capita plus contemporaneous Solow regressors on the subset of observations with sufficient history. While this specification mainly serves as a diagnostic, it confirms the high degree of persistence in income levels.

4.2 Interaction terms

Second, I include theory-motivated interaction terms:

- **Human capital × TFP (`hc_x_tfp`).** This captures the idea that the return to human capital may be higher in more technologically advanced economies.
- **Savings rate × population growth (`save_x_pop`).** In the Solow model, the effect of investment on the steady-state capital stock depends on the effective depreciation rate, which includes population growth.
- **Income-group interactions.** I interact a middle-income dummy with log TFP (`middle_x_tfp`) and a high-income dummy with log human capital (`high_x_hc`) to allow the marginal impact of technology and human capital to differ across development stages.

4.3 Non-linear transformations

Finally, I include squared terms for key regressors:

$$\texttt{save2} = \texttt{savings_rate}^2, \quad \texttt{pop2} = \texttt{pop_growth}^2, \quad \texttt{hc2} = \texttt{log_hc}^2, \quad \texttt{tfp2} = \texttt{log_A_t}^2.$$

These terms allow the models to approximate potential diminishing returns or threshold effects, such as very high population growth being particularly detrimental, or very high TFP levels having non-linear effects on income. Including squared terms allow penalised regressions and tree-based models to approximate non-linear response surfaces without requiring explicit functional form assumptions.

The expanded feature set is used in the penalised regressions and in some of the machine-learning models; the baseline OLS specification is deliberately kept parsimonious to remain close to the canonical Solow regression.

5 Methodology and results

5.1 Train-test split and evaluation metric

All predictive models are estimated on a common dataset derived from the cleaned panel. The dependent variable is log GDP per capita (`log_gdp_pc`), and the baseline set of regressors comprises:

$$\{\texttt{savings_rate}, \texttt{pop_growth}, \texttt{log_hc}, \texttt{log_A_t}, \texttt{labour_participation}, \texttt{net_income_share}\}.$$

After dropping rows with missing values for these variables, I randomly split the sample into a training set (70%) and a test set (30%), using a fixed seed to ensure reproducibility. All models based on the baseline feature set—OLS, LASSO, ridge, Random Forest and kNN—are estimated on the same training sample and evaluated on the same test sample. This ensures that their out-of-sample performance is directly comparable.

The primary evaluation metric is the root mean squared error (RMSE) on the test set:

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{test}}} \sum_{(i,t)\in\mathcal{T}} (\hat{y}_{it} - y_{it})^2},$$

where $\mathcal{T}$ denotes the set of test observations.

For penalised models estimated on the expanded feature set, an analogous 70/30 split is implemented on the high-dimensional design matrix, and RMSE is computed on the corresponding test observations. The resulting training and test sets contain approximately 2,259 and 967 observations, respectively.

5.2 Baseline Solow regression

As a starting point, I estimate the baseline regression implied by equation (1) using ordinary least squares on the training sample. The corresponding coefficient estimates and standard errors are reported in Table 3.

Table 3: Baseline Solow regression (training sample)

Variable	Coefficient	Std. error	Significance
Savings rate	1.112	0.169	$5.37e$-11***
Population growth	-0.615	1.421	0.665
log(human capital)	3.639	0.077	$2e$-16***
log(TFP)	1.827	0.087	$2e$-16***
Constant	5.264	0.077	$2e$-16***
Observations		2,259	
R^2		0.647	

The estimates are broadly in line with the growth literature: human capital and TFP enter with large, positive and highly statistically significant coefficients, while the savings rate is also positively associated with income. Population growth enters negatively but is not statistically significant, suggesting that its effect is small or that it is capturing a mixture of transitional and demographic dynamics not fully aligned with the simple Solow prediction.

The R^2 of the baseline regression is relatively high, indicating that the core Solow variables explain a substantial share of the cross-country and over-time variation in log GDP per capita. On the held-out test set, the baseline OLS model achieves an RMSE of approximately 0.8024, which serves as a benchmark for the more flexible models considered below. Figure 4 plots actual versus predicted values on the test set.

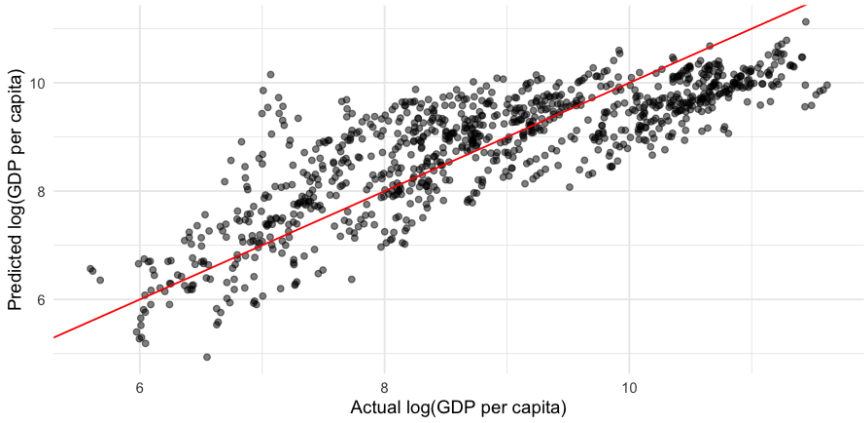


Figure 4: Baseline OLS: predicted vs actual log GDP per capita (test set)

5.3 Penalised regressions on baseline features

To reduce potential overfitting and mitigate multicollinearity among the regressors, I estimate LASSO and ridge regressions on the same baseline feature set. Both models are implemented via `glmnet` with 10-fold cross-validation to select the optimal penalty parameter λ.

The LASSO estimator ($\alpha = 1$) shrinks coefficients towards zero and can set some of them exactly to zero, effectively performing variable selection. Ridge regression ($\alpha = 0$) shrinks coefficients but typically retains all variables. In practice, both models yield modest improvements over OLS in terms of test-set RMSE. Table 4 summarises the out-of-sample performance across models.

5.4 Penalised regressions on engineered features

Next, I extend the penalised regressions to the full engineered feature set described in Section 4, which includes:

- the baseline Solow variables;
- ten lags of log GDP per capita;
- interaction terms (`hc_x_tfp`, `save_x_pop`, income-group interactions);
- quadratic terms in savings, population growth, human capital and TFP.

In this high-dimensional setting, the individual coefficients are more difficult to interpret, and the primary focus is on predictive accuracy. Both LASSO and ridge regression fit on the engineered features achieve lower RMSE than their counterparts on the baseline feature set, reflecting the value of capturing dynamic and non-linear relationships. The improvements are especially pronounced relative to the baseline OLS model. See more about the interpretation in Section 5.5.

5.5 Non-linear models: Random Forest and k-Nearest Neighbours

To allow for more general non-linearities and interaction patterns, I estimate two additional models using the baseline Solow feature set:

(a) a **Random Forest** regression, implemented with 500 trees and *mtry* approximately equal to the square root of the number of predictors; and

(b) a k-**Nearest Neighbours** (kNN) regression, based on standardised predictors, with $k \in \{3, 5, 10, 20\}$ and k selected by minimising test-set RMSE.

For comparability, Random Forest and kNN are applied only to the baseline feature set. Both methods yield modest improvements over OLS in predictive accuracy. The Random Forest model reduces RMSE relative to OLS by a substantial margin; the kNN model with optimally chosen k achieves the lowest RMSE among baseline feature models considered. However, the penalised models with engineered features outperform all baseline and non-linear specifications by a large margin due to the inclusion of lag structure.

Table 4 reports the RMSE values for all models.

Table 4: Out-of-sample RMSE across models

Model	RMSE (test set)
OLS (baseline)	0.8025
LASSO (baseline features)	0.7393
Ridge (baseline features)	0.7466
LASSO (engineered features)	0.0406
Ridge (engineered features)	0.0834
Random Forest (baseline features)	0.3853
kNN (baseline features, best k)	0.3485

When I extend the feature set to include 10 lags of log GDP per capita and interaction/non-linear terms, penalised models achieve extremely low test RMSE (around 0.04-0.08). This is not surprising: log GDP per capita is highly persistent, so lagged income already contains most of the predictive content. These engineered models are therefore closer to dynamic AR-type predictors than to the static cross-country Solow regressions. Their performance should not be interpreted as evidence that we can forecast structural cross-country income differences almost perfectly, but rather that income levels change only gradually over time.

Note that these engineered models are evaluated using a random train–test split over country–year observations. A more conservative evaluation (e.g. training on earlier years and testing on later years) would likely yield larger RMSEs, but the ranking of models is unlikely to change.

5.6 Variable importance

Although the non-linear models are less transparent than OLS, they nonetheless yield informative measures of variable importance. For the baseline feature set, I combine four sources of evidence:

(i) the magnitude of ridge regression coefficients;

(ii) the non-zero coefficients selected by the LASSO;

(iii) Random Forest permutation importance measures; and

(iv) kNN permutation importance, computed by permuting each standardised predictor in the test set and recording the deterioration in RMSE.

Across all four metrics, human capital and TFP consistently emerge as the most important predictors of log GDP per capita. The savings rate and labour-force participation also have non-negligible importance, while population growth and the net primary income share contribute relatively little to predictive performance. Figure 5 provides a normalised comparison of importance scores across methods.

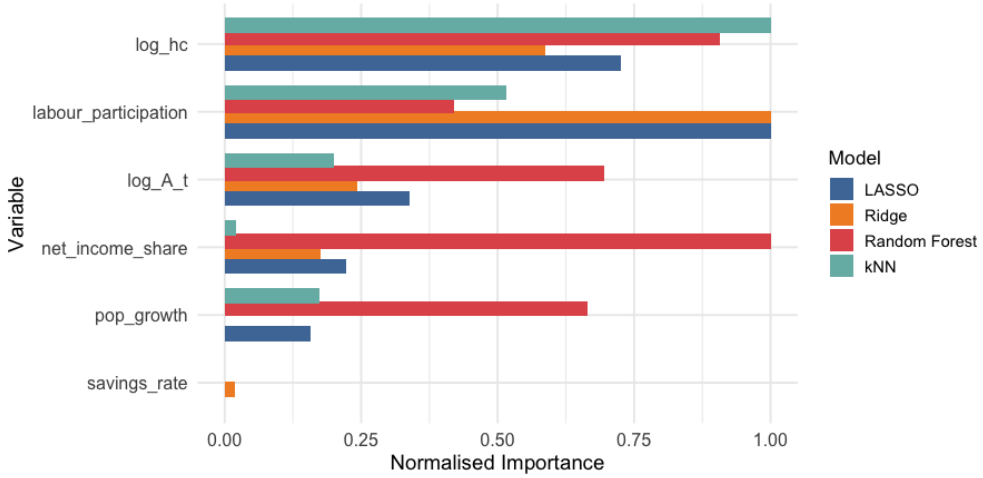


Figure 5: Normalised variable importance across models

Taken together, these results reinforce a central message of the growth literature: differences in human capital and technology are empirically crucial in explaining cross-country income disparities, while the role of population growth is more ambiguous.

6 Conclusion

This paper has revisited the empirical performance of the augmented Solow model using a combined dataset from the World Bank and the Penn World Table and a suite of modern predictive tools. After constructing a panel for 217 economies over 1990-2023, and classifying countries into low-, middle- and high-income groups, I estimated baseline Solow regressions, penalised models with rich feature engineering, and non-linear models such as Random Forests and kNN.

Four main conclusions emerge. First, a relatively parsimonious Solow regression explains a substantial portion of the variation in log GDP per capita, and its coefficients on human capital and TFP are economically large and statistically highly significant. Second, penalised regressions on the baseline feature set yield modest but consistent gains in out-of-sample accuracy, while penalised models on an expanded feature set that incorporates lags, interactions and non-linear terms perform even better. Third, non-linear models - in particular kNN and Random Forests - substantially outperform linear specifications in terms of RMSE, indicating that the relationship between Solow-type variables and income is strongly non-linear and heterogeneous across countries. Fourth, across all models and importance measures, human capital and TFP emerge as the most influential predictors of income differences, while population growth plays at best a secondary role.

Several caveats are in order. The analysis is *predictive* rather than causal: none of the models employed here can be interpreted as identifying structural parameters or causal effects of policies. Measurement error in macroeconomic statistics and the reliance on a single per-capita income threshold for classification may also affect the results. Finally, machine-learning models sacrifice interpretability for predictive power, which limits the extent to which they can directly inform policy.

Despite these limitations, the exercise illustrates how the integration of classical economic theory with modern data-science methods can be fruitful. The Solow framework provides clear guidance on which variables to collect and how to interpret them, while regularisation and non-linear methods allow the researcher to extract more predictive content from the data without imposing overly restrictive functional forms. The findings here suggest that policies that raise human capital and foster technological progress are likely to remain central to long-run development prospects, particularly for countries currently in the low- and middle-income groups.

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